

Math 150A Section 8 and 16
Spring 2025
Exam II
March 13, 2025
Time Limit: 1 Hour 50 Minutes

Name (Print): _____

Student ID: _____

This exam contains 8 pages (including this cover page) and 7 problems. Check to see if any pages are missing. Enter all requested information on the top of this page, and put your initials on the top of every page, in case the pages become separated.

You may *not* use your books or notes on this exam. However, you may use a single, handwritten, one-sided notesheet and a *basic* calculator.

You are required to show your work on each problem on this exam. The following rules apply:

- **Organize your work**, in a reasonably neat and coherent way, in the space provided. Work scattered all over the page without a clear ordering will receive very little credit.
- **Mysterious or unsupported answers will not receive full credit.** A correct answer, unsupported by calculations, explanation, or algebraic work will receive no credit; an incorrect answer supported by substantially correct calculations and explanations might still receive partial credit. This especially applies to limit calculations.
- If you need more space, use the back of the pages; clearly indicate when you have done this.
- **Box Your Answer** where appropriate, in order to clearly indicate what you consider the answer to the question to be.

Problem	Points	Score
1	10	
2	10	
3	10	
4	10	
5	10	
6	10	
7	10	
Total:	70	

Do not write in the table to the right.

1. (10 points) **TRUE or FALSE!** Write

(a) every inflection point of $f(x)$ is also a critical point of $f(x)$

(b) the function $f(x) = x^{1/3}$ has no critical points

(c) if $f(x)$ has three different x -intercepts, then $f'(x)$ must have at least two x -intercepts

(d) every function has an absolute maximum

(e) if $f''(3) = -2$, then $f(x)$ has a local maximum at $x = 3$

2. (10 points)

We want to construct a cylindrical can with a bottom but no top that will have a volume of 30 cm³. Determine the dimensions of the can that will minimize the amount of material needed to construct the can.

Hint: the volume of a cylinder is $\pi r^2 h$, the circumference of a circle is $2\pi r$ and the area of a circle is πr^2 .

3. (10 points) Consider the function

$$g(t) = t^5 - 5t^4 + 8$$

- .
- (a) Identify the critical points of the function
 - (b) Determine the intervals on which the function increases and decreases.
 - (c) Classify the critical points as relative maximums, relative minimums or neither.
 - (d) Determine the intervals on which the function is concave up and concave down.
 - (e) Determine the inflection points of the function.
 - (f) Use the information from steps (a) (e) to sketch the graph of the function.

4. (10 points) Use L'Hospital's Rule to calculate the following limits:

(a)

$$\lim_{x \rightarrow 0} \frac{1}{x^2} - \frac{\cot(x)}{x}.$$

(b)

$$\lim_{x \rightarrow \infty} x^{1/\ln(x)}.$$

5. (10 points) Consider the equation

$$x^9 + 7x^5 + x^5 + 2x^3 + 3x + 4 = 0.$$

- (a) State the Mean Value Theorem

- (b) Use the Intermediate Value Theorem to show that the above equation has at least one solution

- (c) Use the Mean Value Theorem to explain why the equation has no more than one solution

6. (10 points) For each of the following, provide an example or explain why it does not exist.

(a) a function with no critical points

(b) a function with inflection points at -1 and 1

(c) a nonconstant function $f(x)$ whose derivative is always zero

(d) a function with a global maximum at $x = 1$

(e) a critical point of the function $3 - |x - 3|$

7. (10 points) Find the critical points of each of the following functions. Classify them as local maxima, local minima, or neither.

(a) $2x^3 + 9x^2 + 3x + 4$

(b) $9x^{5/3} - 25x^{3/5}$